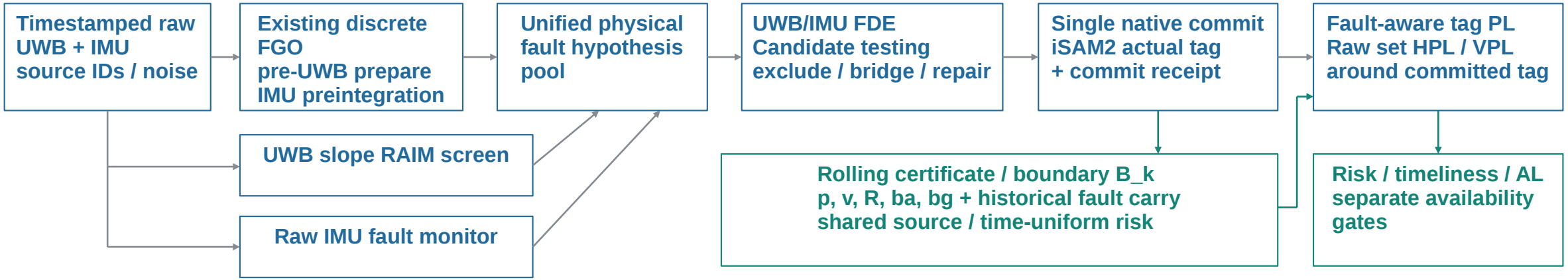
The System Today: One FGO, Joint Fault Handling, Rolling Integrity

How are estimation, fault exclusion and protection connected in the live UWB–IMU system?

20 Hz UWB / 200 Hz IMU • default LF/LI = 30/30 epochs • committed tag-position PL



T-SIM: conditional raw integrity | T-REAL: engineering FDE only; raw unqualified



B_k → B_(k+1): state/error, possible faults and source/risk survive; risk horizon ≠ 30 epochs. No second FGO or all-history re-solve.

Raw measurements drive one incremental FGO and parallel fault monitors; the selector commits at most one native update. A rolling certificate carries state uncertainty, fault consequences and source/risk identity across marginalization.

Takeaway: A coupled incremental estimator and rolling fault-aware position integrity layer goes beyond a per-epoch UWB RAIM test.

Source: SC8_RESULTS, Architecture; frozen solver 61907ab; conditional SIM / empirical REAL separated

E00 — What Was Actually Tested, and Which Comparisons Are Valid?

Which datasets, ground truths and algorithm modes support a defensible comparison?

143 registered runs: 130 SUCCESS | 12 RESOURCE_LIMIT | 1 FAIL; 159 attempts + 1 ROS control

T-SIM

Registered physical law → O-FULL / O-NOFDE

→ Known truth; conditional PL

T-REAL

Original recordings → O-FDE-ENG / O-ENG-NOFDE

→ CAL disclosure; raw unqualified

T-SEMI

Real + fixed overlay → Same engineering paths

→ Injected label only; no raw PL

ID	Recording	Duration	Physical reference / boundary
R1	ISAS-Walk2	76.09 s	tracker→IMU/tag OPEN; proxy
R2	ISAS-Walk3	81.66 s	same reference OPEN
R3	Own Vicon, no obstacle	73.39 s	GT-assisted segmented CAL/TEST
R4	Own Vicon, obstacle	11.39 s	GT-assisted; short recording
R5	STAR loop-3d-z_s3	150.10 s	gyro/rig input convention OPEN
R6	STAR loop-2d-fast_s3	55.40 s	same input concern

SFUISE: official rejection ON; ONLINE causal; FINAL same-run revised history.

Rows identify pre-registered real recordings, rather than six successful examples. The producer-to-estimator lanes keep conditional simulation integrity separate from empirical real data and overlays.

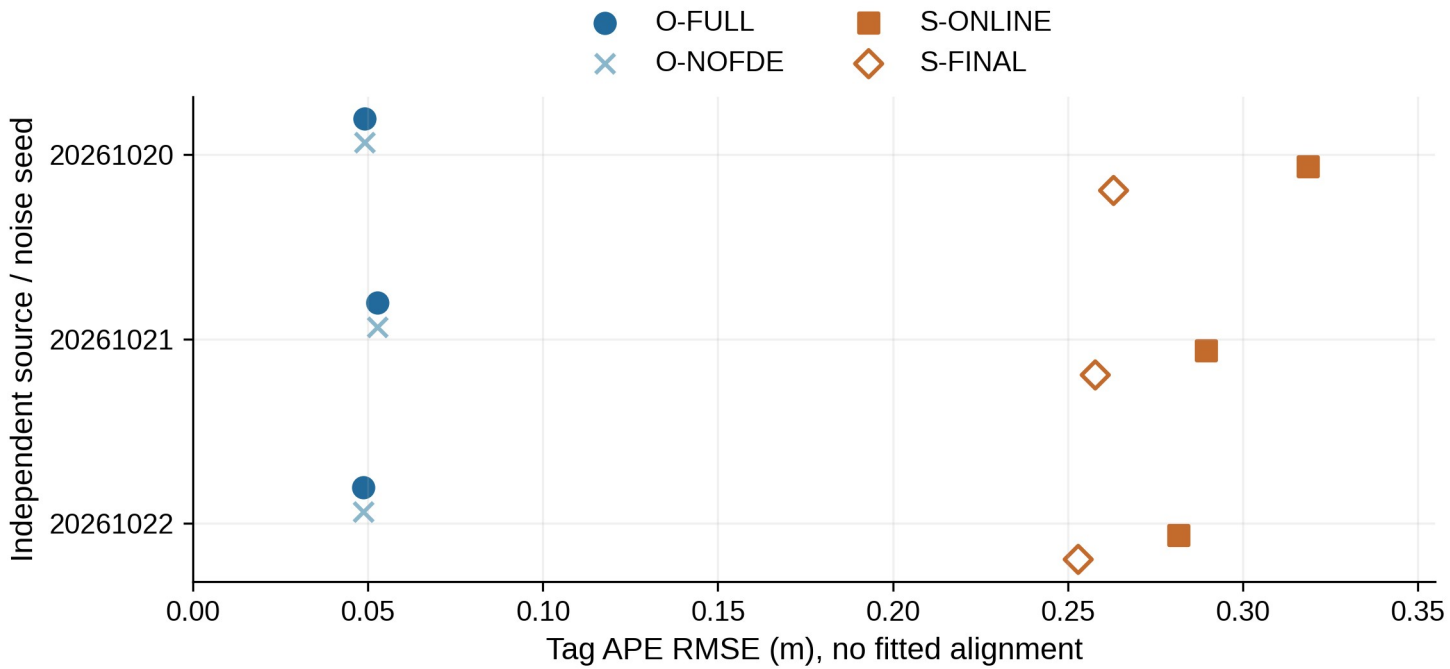
Takeaway: Comparability comes before APE rankings: Walk reference and STAR input contracts remain open.

Source: SC8_RESULTS, E00; frozen solver 61907ab; conditional SIM / empirical REAL separated

E01 — Healthy Simulation: Good Accuracy Under the Declared Model

How accurately does the coupled pipeline localize without imposed faults?

O-FULL tag APE RMSE 0.0487176–0.0526469 m; default timing / timely-spatial profile passes 3/3



Full path / model	Observed result
Tag APE RMSE (m)	0.0487176–0.0526469
Body ARE RMSE	0.0778–0.1454 rad
Seed20 core / sink P95	23.73 / 23.89 ms
Study limits	40 / 50 ms; timely ≥95%

Per-run output-stamp scoring.
FULL / NOFDE: 600 states each.
SFUISE: 300 online; 298 valid final.
Final history is not an online gain.

Each point is one independent seed; lower world-frame tag APE is better, without trajectory alignment. SFUISE uses sensor-only initialization, while ours receives an independently noisy initial observation: this is not an optimizer-only comparison.

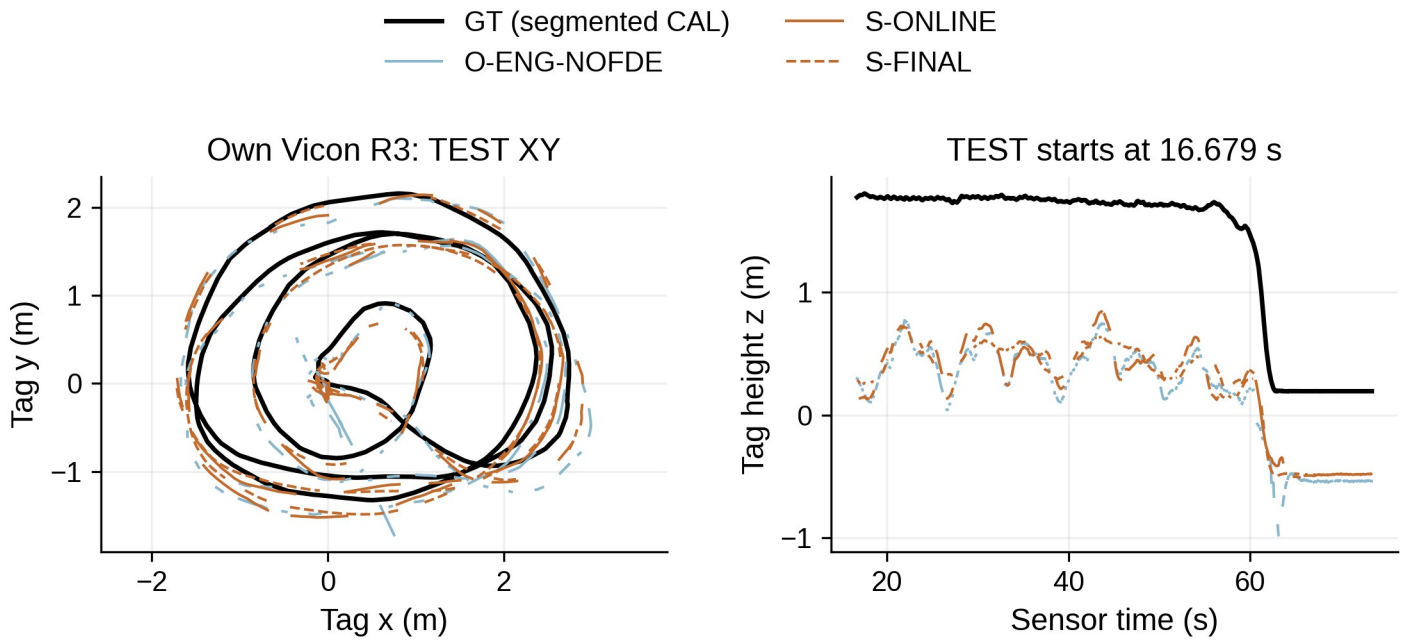
Takeaway: The healthy pipeline delivers approximately 5 cm tag RMSE in the registered simulated model.

Source: SC8_RESULTS, E01; frozen solver 61907ab; T-SIM: conditional tag-position PL

E02 — Real-Data Validation Is Currently an Interface Problem

Which real-data comparisons can we physically interpret?

Walk physical APE / ARE / RPE = NA; STAR input validation OPEN; own Vicon is GT-assisted



Family	Usable evidence	Open boundary
Walk R1/R2	Time coverage; proxy diagnostic	Tracker–body/tag reference
Own R3/R4	Empirical TEST; CAL disclosed	TEST-involved fit; not blind
STAR R5/R6	Frozen-interface behavior	Gyro/rig convention; no intrinsic ranking

R3 FULL: no complete TEST trajectory.
STAR R5 S-ONLINE APE: 15.85 km.
Input OPEN: no intrinsic-method ranking.

Traces show the frozen own-Vicon TEST output and height reference, with its GT-assisted calibration disclosed. The matrix separates physically unresolved Walk references and STAR input conventions from conditional empirical scoring.

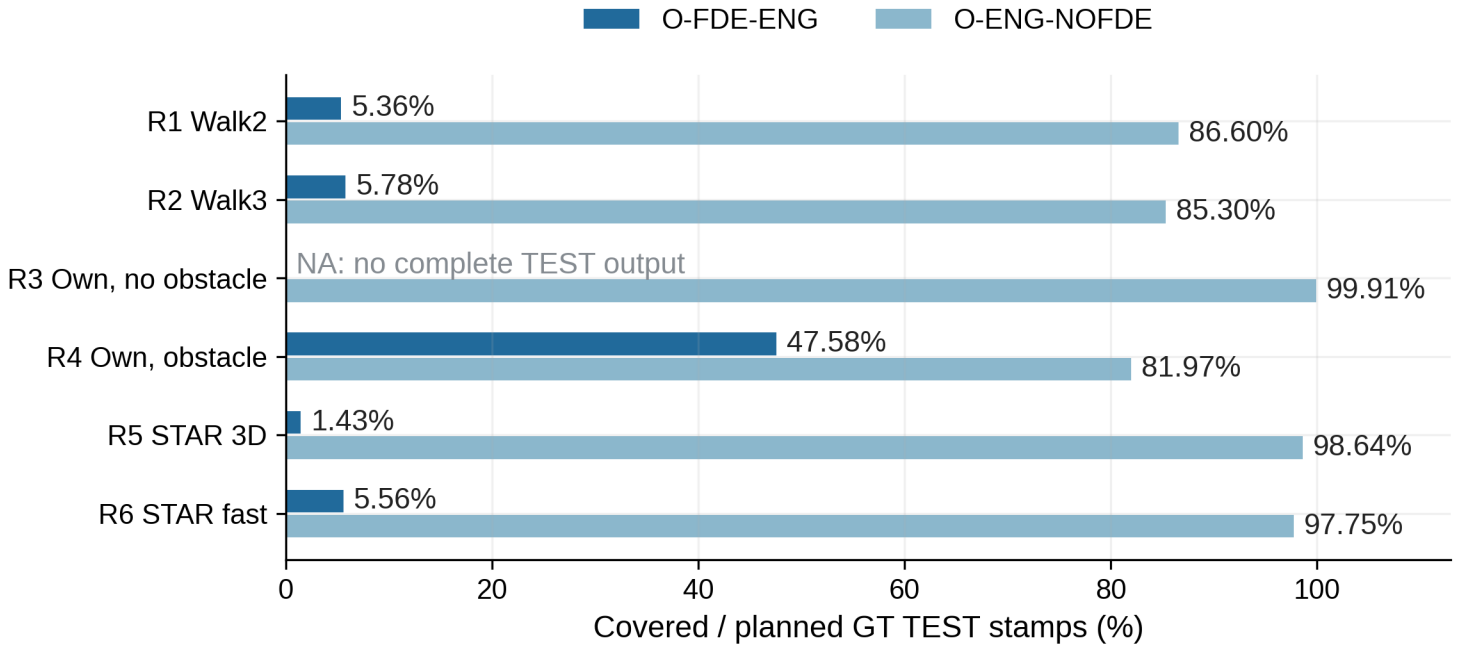
Takeaway: Apparent algorithm failures cannot yet be separated from missing physical frame and timing validation.

Source: SC8_RESULTS, E02; frozen solver 61907ab; T-REAL: empirical, raw unqualified

E02 — Real FDE Execution, Not Nominal Localization, Is the Bottleneck

Can the fault-handling pipeline sustain a complete real recording?

4/6 O-FDE-ENG real runs hit the resource limit; every real raw PL remains unqualified



GT grid	FDE covered/N	NoFDE covered/N
R1	204/3805	3295/3805
R2	236/4083	3483/4083
R3	NA	16602/16617
R4	1591/3344	2741/3344
R5	505/35342	34861/35342
R6	723/12993	12701/12993

Walk: temporal proxy grid only.
R3 FULL has no saved TEST output.
Rates differ from commit/attempt rate.

Bars show coverage of the registered GT TEST grid, with interpolation limited to 100 ms; NA is not zero. This measures emitted-output continuity, not accuracy or raw integrity, and Walk coverage is temporal only.

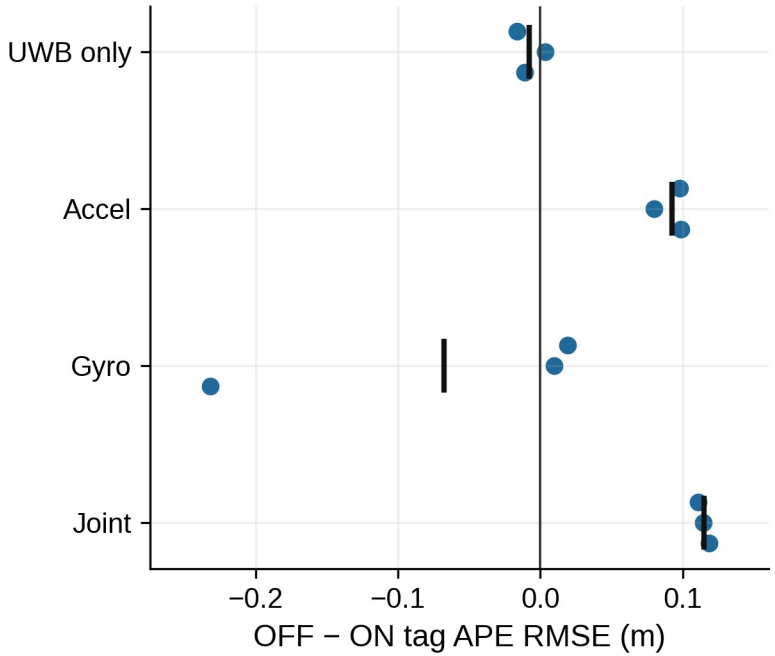
Takeaway: The main real-data blocker is FDE execution continuity, beyond nominal FGO data loading.

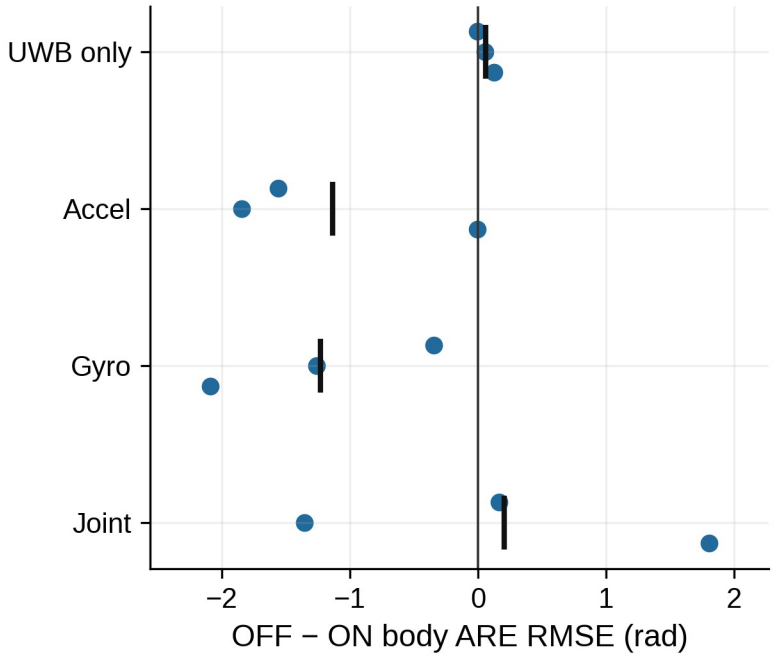
Source: SC8_RESULTS, E02; frozen solver 61907ab; T-REAL: empirical, raw unqualified

E03 — FDE Benefits Are Fault-Dependent, Not Uniformly Positive

Does FDE actually improve position and attitude accuracy?

OFF − ON: positive is improvement; negative is degradation; all three seeds remain visible





Fault	ΔAPE (m)	ΔARE (rad)
UWB	-0.007908	+0.05806
Accel	+0.092363	-1.13562
Gyro	-0.067733	-1.23141
Joint	+0.114792	+0.20435

Black ticks are seed means.
Accel: better position, worse attitude.
Gyro: both means deteriorate.
Joint: attitude gain varies by seed.

Each point is a matched seed; positive OFF−ON means FDE reduced RMSE, while negative values show degradation. Position improvement cannot cancel worsening body attitude, particularly under accel or gyro faults.

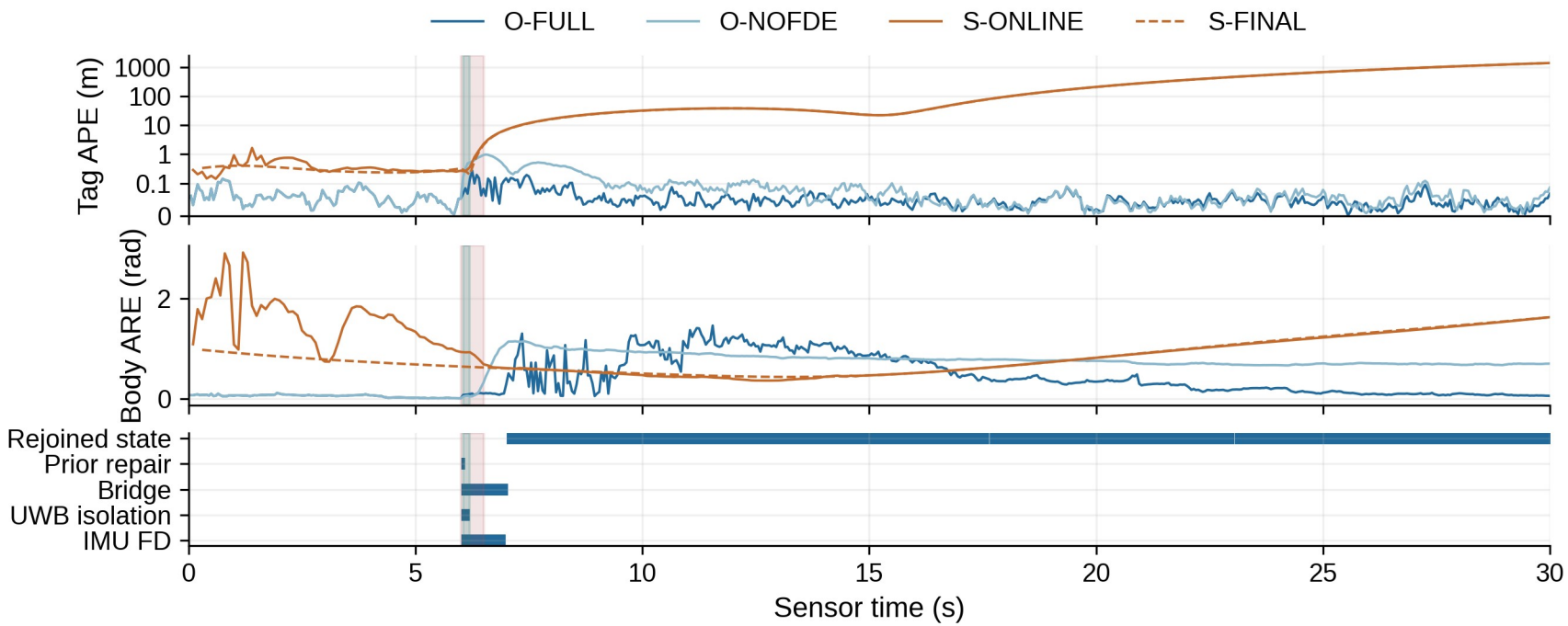
Takeaway: Accel/joint positioning improves, but gyro-only accuracy and state recovery remain unresolved.

Source: SC8_RESULTS, E03; frozen solver 61907ab; T-SIM: conditional tag-position PL

E03 — Real Isolation Can Still Be Followed by Estimation Drift

What happens before, during and after a detected joint fault?

T-SIM SC8-J / seed 20261020: actual prior repair + bridge + rejoined receipts; accuracy is separate



Source / event	Magnitude / receipt
UWB bias	+2.25 m / 3 epochs
Raw accel	+20 m/s² / 500 ms
First FD / bridge	Epoch 121 / 121
First rejoin4	Epochs 141–144

Burgundy: accel pulse.
Teal: UWB pulse.
Rejoined is a persistent state, not an accuracy certificate.

Bands mark source-level injections and receipt rows mark actual detection, isolation, prior repair and rejoined state. The APE axis is linear below 0.1 m and logarithmic above; a successful transaction does not certify state-error recovery.

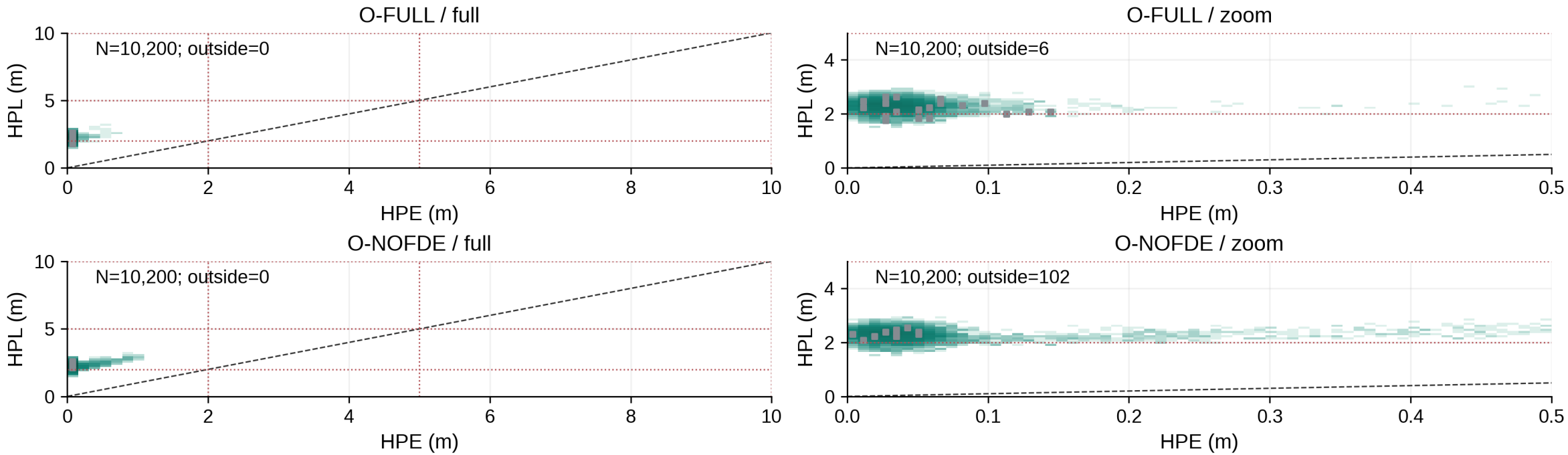
Takeaway: Post-FDE state consistency and recovery determine net accuracy; isolation alone is insufficient.

Source: SC8_RESULTS, E03; frozen solver 61907ab; T-SIM: conditional tag-position PL

E04 — Horizontal Stanford: Conservative Bounds, Conditional Coverage

How do horizontal position errors relate to the emitted HPL?

77,131 qualified T-SIM receipts overall; observed H/V underbounds = 0; correlated samples ≠ risk trials



HPE is horizontal and HPL vertical; darker teal log-count bins have more timely samples, and points above dashed y=x are empirically covered. Gray bins are late (excluded from timely availability); dotted lines mark research ALs (2/5/10 m), and zero correlated violations do not establish empirical 10^-6 risk.

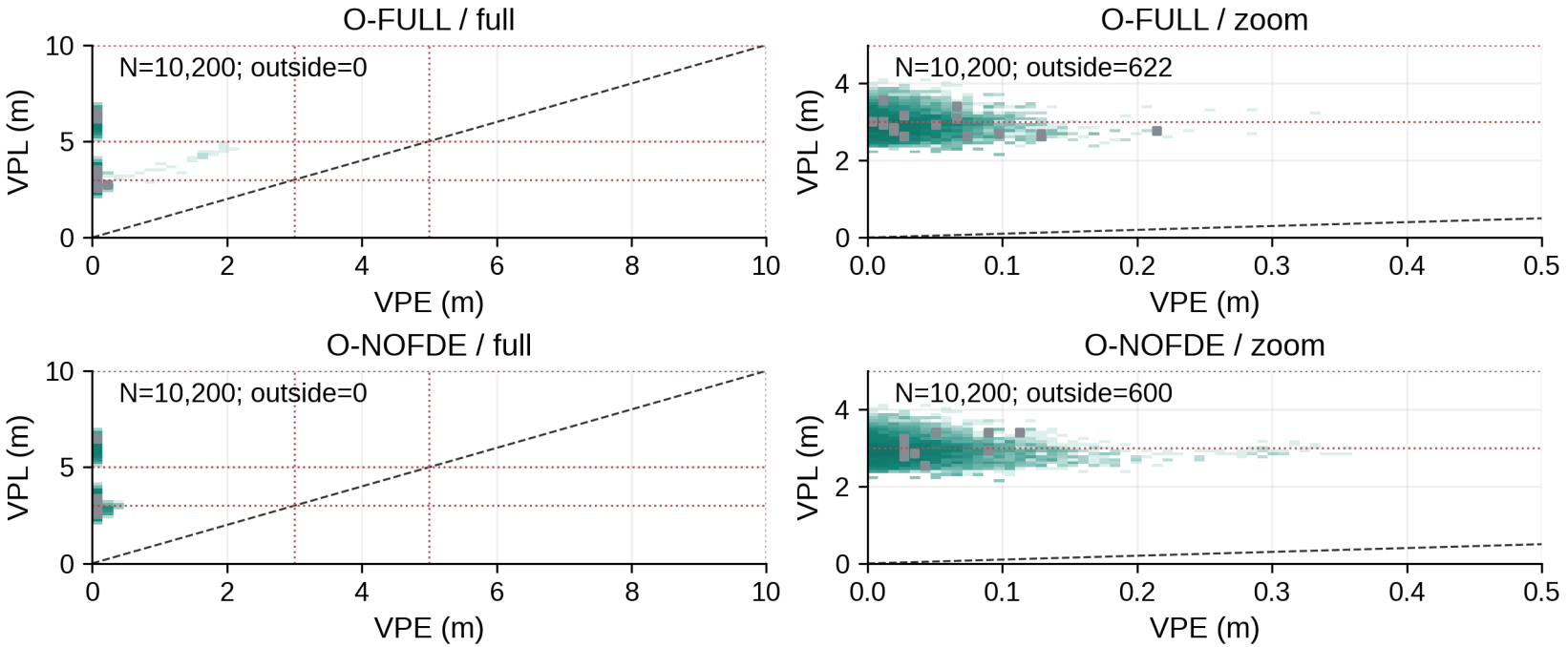
Takeaway: Horizontal bounds cover the registered errors but remain much wider than the observed error.

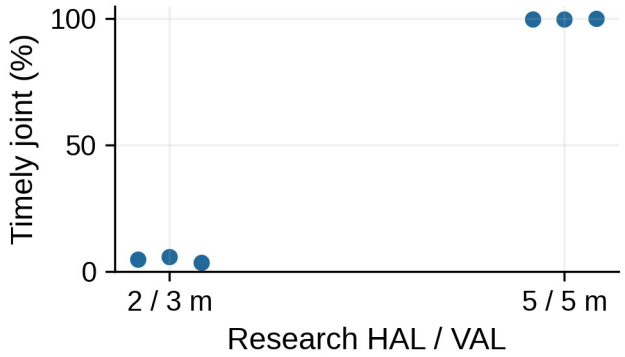
Source: SC8_RESULTS, E04; frozen solver 61907ab; T-SIM: conditional tag-position PL

E04 — Vertical Stanford: Availability Depends on the Application AL

Why can a centimeter-level estimate still be unavailable?

Same estimates and PL: research AL (2,3) m gives 3.5–5.667%; (5,5) m gives 99.67–100%





HAL / VAL	Joint timely
2/3 m	3.5–5.667%
5/5 m	99.67–100%

O-FULL; 3 independent seeds.
Denominator: all 600 epochs / seed.
No application certification.

VPE is horizontal and VPL vertical; darker teal bins contain more timely samples, gray bins are late, and bins above dashed y=x have VPL≥VPE. Dotted research ALs (3/5/10 m) test admissibility; the healthy AL comparison changes evaluation only, not estimates or PL, and is not certification.

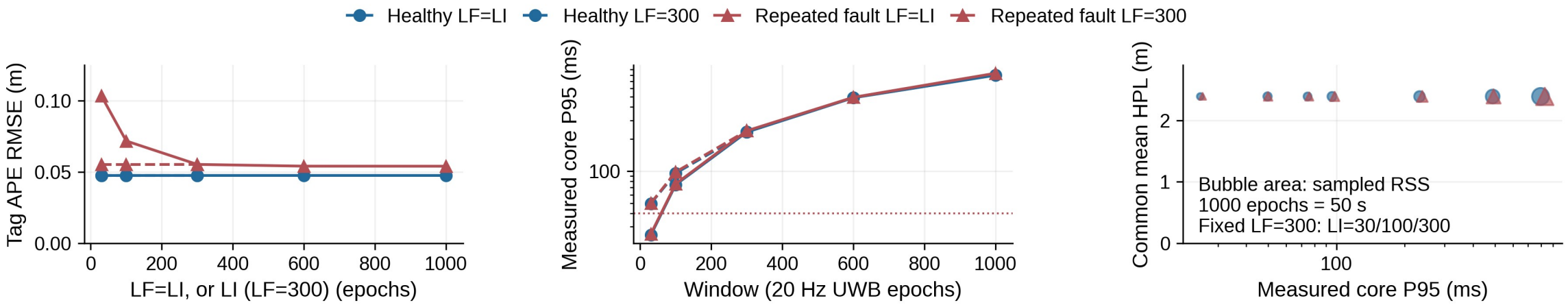
Takeaway: PL tightness and application availability are distinct from estimator accuracy, especially vertically.

Source: SC8_RESULTS, E04; frozen solver 61907ab; T-SIM: conditional tag-position PL

E05 — 30 vs 1000 Epochs: When Is a Longer Window Worth It?

Does retaining much more history improve accuracy or integrity enough to justify the cost?

7 unique configurations × 2 streams; common epochs 1101–2100; default core study target = 40 ms



Same common interval	30/30	1000/1000	Interpretation
Healthy APE (m)	0.04753682	0.04754147	No meaningful gain
Healthy core P95 (ms)	25.096	795.501	Higher measured cost
Fault APE (m)	0.10345833	0.05404642	Benefit in this stream

APE uses the same complete 1,000-epoch scoring interval; bubble area denotes sampled RSS, not process peak. LF, LI and recovery replay vary together in the coupled sweep, so its fault benefit cannot be attributed to PL alone.

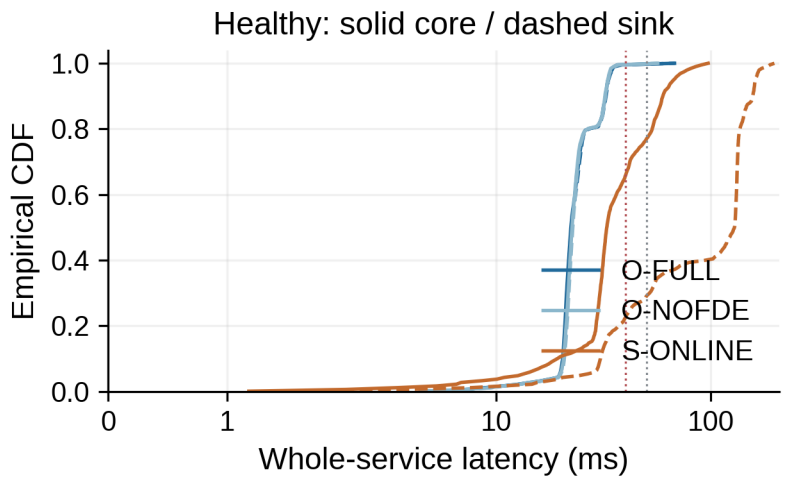
Takeaway: Keep 30/30 as the online reference; extend history only when measured recovery benefits justify cost.

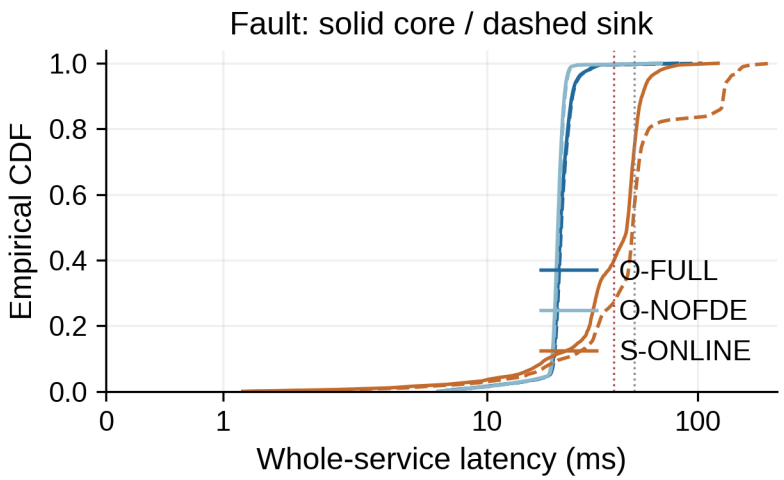
Source: SC8_RESULTS, E05; frozen solver 61907ab; T-SIM: conditional tag-position PL

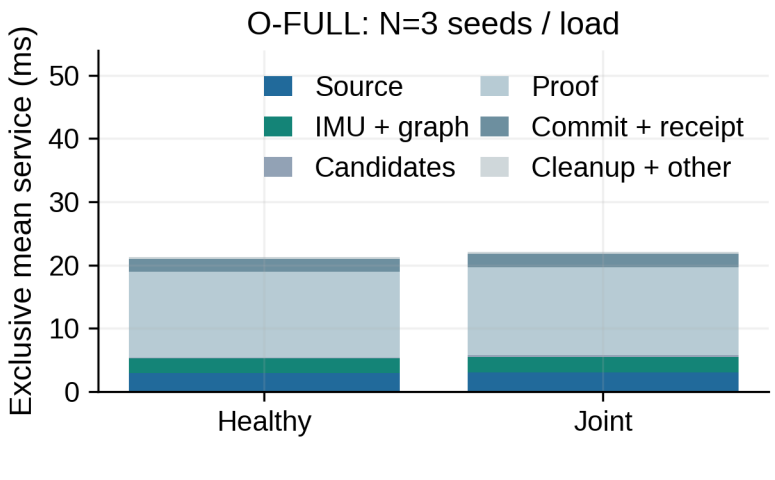
E06 — Runtime Breakdown: Fast Simulation, Expensive Real FDE

Where does the computation go, and which timing numbers are comparable?

Core P95 ≤40 ms + sink P95 ≤50 ms + timely spatial ≥95%; two E06 serial confirmations pass







Run / capability	Core P95 (ms)	Sink P95 (ms)	Boundary
E01 H / seed20 FULL	23.726	23.893	Full service
E06 J / seed20 FULL	24.893	25.162	Full service
E01 H / seed20 NOM	2.100	2.303	Nominal only; no matched PL
E02 real O-FDE-ENG	up to 13099.0	up to 1.14e+06	Measured prefix; 4/6 resource-limited

Curves show measured whole-service latency; solid is core and dashed is sink, while stacked bars contain additive exclusive means. Stage P95s are not additive, and SFUISE buffering/software-clock timing represents different capabilities.

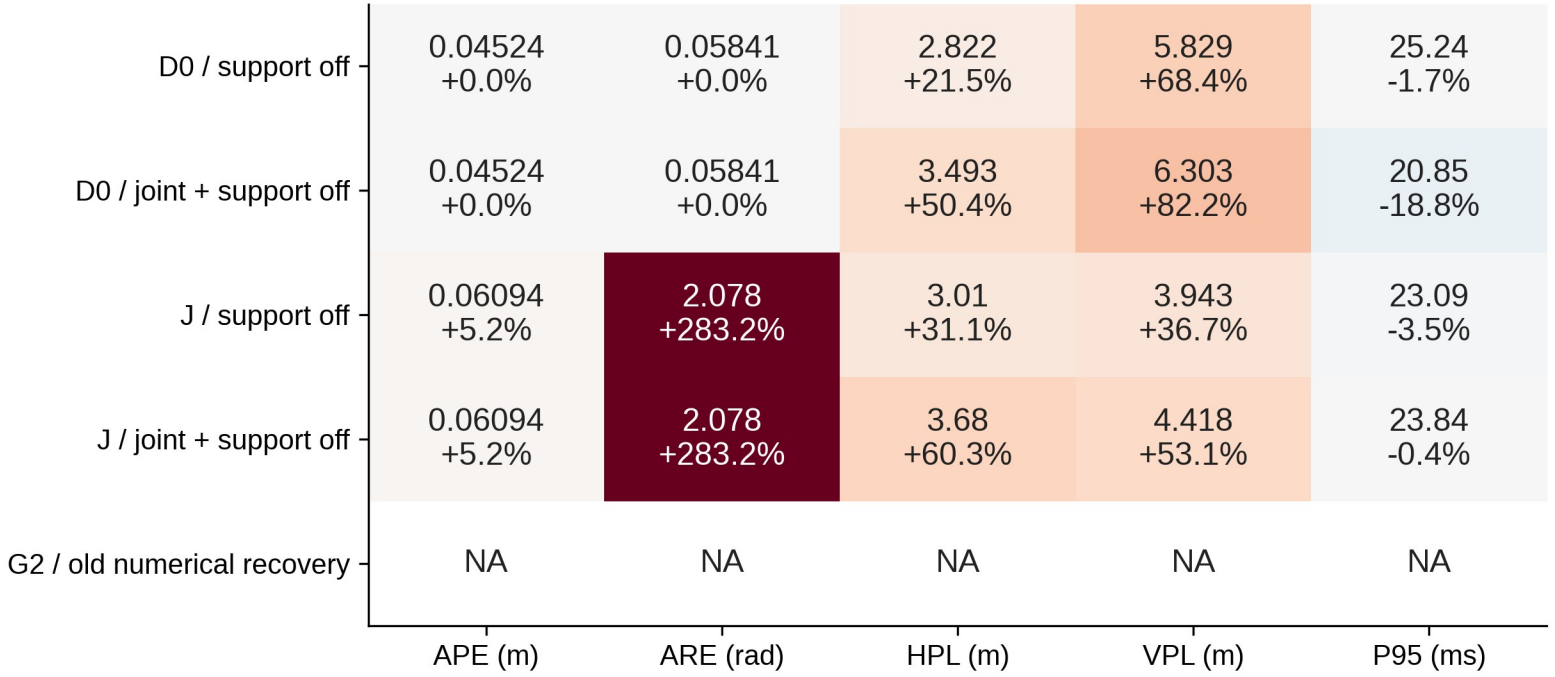
Takeaway: The small-window simulated full path meets study P95 targets; current real engineering FDE does not.

Source: SC8_RESULTS, E06; frozen solver 61907ab; conditional SIM / empirical REAL separated

E07 — Ablations: Which Integrity Components Actually Matter?

Are rolling constraints and recovery logic causally necessary, or just complexity?

Removing support/joint constraints widens PL; the old G2 numerical candidate commits 0/80



M — same center

Old/new PL at one committed tag center: a tightening witness.

A — full trajectory

Changing fault support can change the subsequent state trajectory.

P — performance

Whole-service measurements; do not sum module percentiles.

G2: 0/80; epoch1 abort = 4.874 ms.
APE / PL: NA; no common valid baseline.
Joint + support is a combined switch.

Cells show actual values and signed changes from matched FULL; red means increase and blue means decrease. G2 has no qualified output, so APE/PL are NA; the joint/sample0 switch also removes support and cannot isolate both effects.

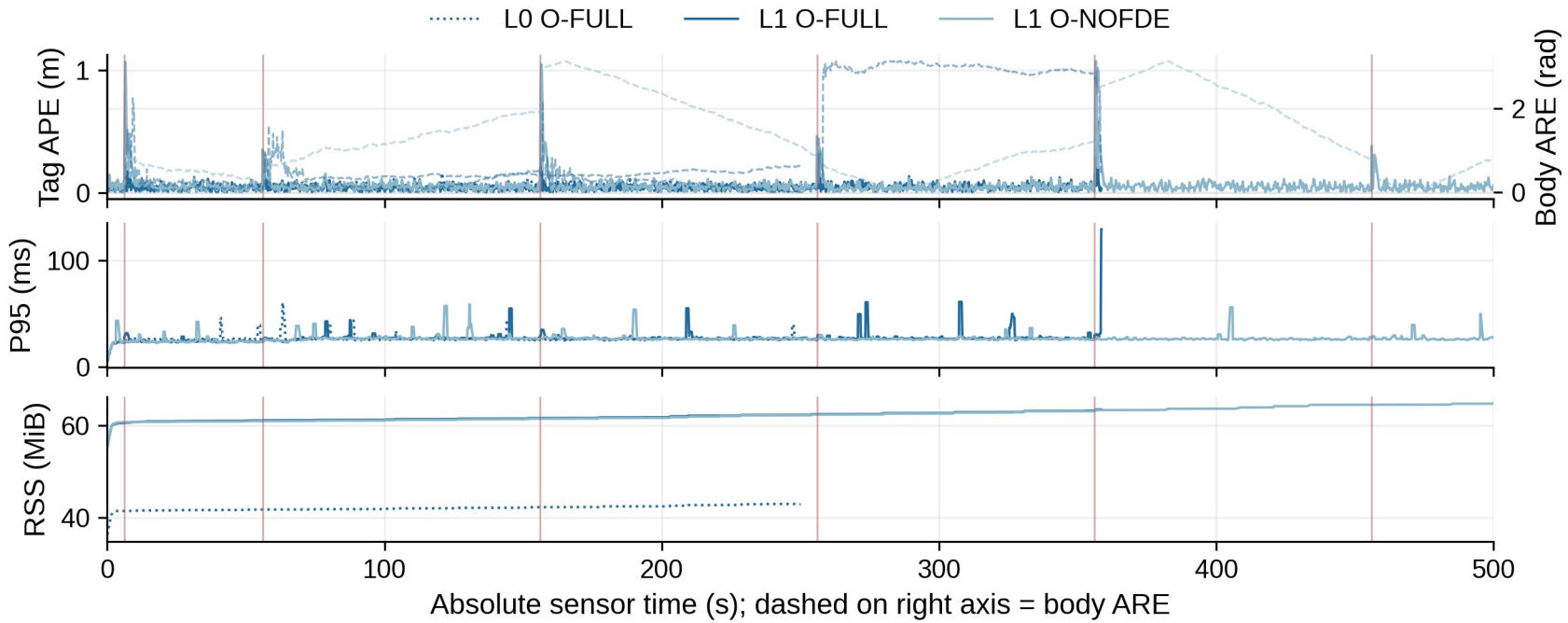
Takeaway: Set constraints and geometry-aware numerical recovery have measurable effects; failures remain visible.

Source: SC8_RESULTS, E07; frozen solver 61907ab; T-SIM: conditional tag-position PL

E08 — Long-Run Integrity: Healthy Complete, Repeated-Fault Case Censored

Does the full pipeline sustain 10,000 epochs and process the last injected fault?

All three processes exit SUCCESS; actual FULL L1 coverage is 7171/10000, not full completion



Run	Committed / plan	Queue peak
L0 FULL	5000/5000	13
L1 FULL	7171/10000	80
L1 NOFDE	10000/10000	15

FULL L1: 2,829 refused epochs.
The final scheduled fault is unverified.
Slots grow; bounded active window does not mean constant total memory.

Curves stop at actual committed output; a blank FULL suffix is unprocessed data, not zero error. Vertical marks are scheduled faults, and the final episode beyond epoch 7171 was not verified by FULL.

Takeaway: Continuous recovery at scale is unproven: FULL stops where the no-FDE route continues.

Source: SC8_RESULTS, E08; frozen solver 61907ab; T-SIM: conditional tag-position PL

Synthesis — What We Have Demonstrated, and What Remains Open

Which claims can we defend in a paper today?

Demonstrated: conditional method + measured simulation results; not hardware-qualified real integrity

Defensible claim	Direct SC8 evidence	Boundary / not established
Conditional rolling position PL	77,131 qualified receipts; 0 observed underbounds	Registered source/noise/fault law only; correlated observations
Healthy tag positioning ≈5 cm	3 fixed independent simulation seeds	Different initialization information from SFUISE
FDE benefit is fault-specific	Accel/joint position gains; gyro regression	Attitude / late recovery can deteriorate
Real comparison remains incomplete	Walk/STAR interface gaps; real FDE censoring	Engineering empirical results; no raw real qualification
Default small window justified	25.096 → 795.501 ms healthy core P95	Longer lag helps this fault stream; not generally real-time

Additional gaps: SFUISE velocity/bias export; invalid history prefix; weak-UWB and S1/S2 studies omitted.

Each row pairs a defensible claim with direct evidence and its limiting assumptions. Mathematical containment, empirical error, execution continuity and application certification answer different questions.

Takeaway: The strongest current contribution is conditional rolling fault-aware position integrity; real execution remains open.

Source: SC8_RESULTS, Synthesis; frozen solver 61907ab; conditional SIM / empirical REAL separated

Questions for the Advisor — Decisions Needed Before the Next Study

Which scope, resources and priorities should guide the next study?

Proposed next action: close independent calibration → stabilize real FDE → limited newly frozen rerun

1. Publication scope

Should the near-term paper focus on the conditional rolling raw position PL + joint FDE architecture, with honest execution/real-data limitations, or is physically qualified real-data integrity required before submission?

Decision: theory + qualified simulation first, or hardware validation first.

2. Independent calibration resources

Can we obtain independent tracker↔IMU/body↔tag 6-DoF extrinsics, timing and anchor-world references, or collect a new calibration sequence with three-axis motion and independent held-out TEST?

Decision: identify the hardware/protocol owner and available sensors.

3. State recovery or execution continuity

Should we prioritize explaining and fixing gyro-induced attitude degradation / long-term state recovery, or first make engineering FDE process complete real recordings within a bounded resource budget?

Decision: algorithmic research priority or engineering qualification priority.

4. Application integrity requirements

For the intended deployment, what HAL / VAL, TTA, covered fault classes and failure action should determine meaningful protected availability?

Decision: target application and first defensible safety requirements.

The four questions request concrete decisions on scope, calibration resources, recovery priorities and application requirements. The next protocol should follow those decisions before a limited, newly frozen rerun.

Takeaway: Requested outcome: agree on the paper scope, calibration resource owner and research priority order.

Source: SC8_RESULTS, Decisions; frozen solver 61907ab; conditional SIM / empirical REAL separated